Chap 2 电化学中重要的概念

一. Gibbs energy $G = H - TS = U + pV - TS$

1. 变形

① closed system
② non-expansion work
③ constant composition

$\left.\right\} dG = Vdp - SdT$

将 $V_m = \frac{RT}{p}$ 代入得 $G_m(p) = G_m^{\ominus} + RT\ln\frac{p}{p^{\ominus}}$

2. 对真实气体：$p \to f$ 逸度 fugacity. $f = \Phi p$

effective pressure：$G_m = G_m^{\ominus} + RT\ln\frac{f}{p^{\ominus}}$.

fugacity coefficient Φ：$\ln\Phi = \int_0^p \frac{Z-1}{p}dp$

Compression factor Z 表征偏差程度

二. 化学势

1. 对于混合溶液中的组分：Partial molar Gibbs energy

$\mu_J = \left(\frac{\partial G}{\partial n_J}\right)_{p,T,n'} \xrightarrow[G = n_J G_{J,m}]{\text{真实气体}} G_{J,m} = \mu^{\ominus} + RT\ln\frac{p}{p^{\ominus}}$

∴ real gas：$\mu = \mu^{\ominus} + RT\ln\frac{\Phi p}{p^{\ominus}} = \mu^{\ominus} + RT\ln\frac{f}{p^{\ominus}}$

2. Raoult's law and ideal solution 摩尔分数

$p_A = x_A p_A^*$：一定T的稀溶液中，溶剂蒸气压 = 纯溶剂的x蒸气

3. Henry's law and ideal-dilute solutions

$p_B = x_B K_B$：一定T密封容器中，气体分压 ∝ 溶液中摩尔浓度

4. 对溶液 solution 中的溶剂 solvent

$\mu_A = \mu_A^{\ominus} + RT\ln p_A = \mu_A^* - RT\ln p_A^* + RT\ln p_A$

纯：+ *

有效浓度 $= \mu_A^* + RT\ln\frac{p_A}{p_A^*} = \mu_A^* + RT\ln x_A$

· 真实溶液中，$x_A \to a$ 活度 $a_A = \frac{p_A}{p_A^*} = \mu_A^* + RT\ln a_A$

活度系数 γ：$a_A = \gamma_A x_A$. $\gamma_A \to 1$ as $x_A \to 1$

5. 稀溶液中的溶质 solute 某组分偏离理想溶液的程度

$\mu_B = \mu_B^* + RT\ln\frac{p_B}{p_B^*} = \underbrace{\mu_B^* + RT\ln\frac{K_B}{p_B^*}}_{\Downarrow\ \mu_B^{\ominus}} + RT\ln x_B$

∴ $\mu_B = \mu_B^{\ominus} + RT\ln x_B$

x 活度 $a_B = \frac{p_B}{K_B} = \gamma_B x_B \Rightarrow \mu_B = \mu_B^{\ominus} + RT\ln a_B$

三. 化学势的应用

1. 反应程度 ξ $d\xi = \frac{dn_i}{\nu_i} \to$ stoichiometric number

2. 化学计量比 $\nu_i = \frac{\Delta n_i}{\Delta \xi}$

$dG = \sum_i \mu_i dn_i = \sum_i \nu_i \mu_i d\xi$

3. 反应系数 quotient $Q_r = \prod_i a_i^{\nu_i}$

平衡时，$K = Q_r$. $RT\ln K = -\Delta_r G^{\ominus} = nFE^{\ominus}$

4. 反应 Gibbs energy $\Delta_r G = \left(\frac{\partial G}{\partial \xi}\right)_{p,T} = \sum_i \nu_i \mu_i = \Delta_r G^{\ominus} + RT\ln Q_r$

Chap 2 Electrolyte Solution

一. electrolysis 电解

1. Faraday's Law：$m = \left(\frac{Q}{F}\right)\left(\frac{M}{n}\right)$

m ← 电极析出物质的质量；M ← 摩尔质量；n ← 化合价(abs)

电化当量 $N = \frac{Q}{nF}$. Rate(mol/s) $= \frac{dN}{dt} = \frac{i}{nF}$.

二. Electrolyte solution 电解质溶液

1. Electrolyte：溶剂与溶质分子间相互作用，使各组分分离

2. Solvation 溶解

3. Lonization 离子化：原子或分子通过得失电子而获得正负电荷的过程，常与其它反应一并发生.

- Hydration 水化：分子与水结合的过程，常可逆
- Hydration Shell：水分子或多或少形成的团围绕着水分子，在一定程度上与水分子吸引的其它物质相连. 亲水.
- 水化数：与化合物结合的平均分子数 > 与其它水分子结合

4. 分类 $\left\{\begin{array}{l}\text{concentrated}\\ \text{diluted: low 浓度}\end{array}\right.$ $\left.\begin{array}{l}\text{strong: electrolyte}\\ \text{weak}\end{array}\right\}$ 描述物质中电荷

5. Conductance 电导 $G = \frac{1}{R} = \kappa\frac{A}{l}$ ← 流动难易程度

Conductivity 电导率 $\kappa = \frac{1}{\rho} = F\sum_i |z_i| u_i c_i$

- Migration 电迁移.
- Mobility 淌度：单位场强 (V/cm) 下离子迁移速度

$u_i = \frac{v}{\xi} = \frac{|z_i|e}{6\pi\eta r}$.

- 电迁通量 $J = c \cdot v$. 电流密度 $j = zFJ = \kappa \cdot E$

$\Rightarrow \kappa = \frac{zFc \cdot v}{E} = zFcu$

- Molar ionic conductivity $\Lambda_m = \frac{\kappa}{c}$ $S\,cm^2/eq$

当量电导率 $\Lambda^{eq} = \frac{\kappa}{C_{eq}} = \frac{\kappa}{|z|c} = \frac{\Lambda}{|z|} = F(u_+ + u_-)$

∴ $\Lambda^{eq}(\lambda) = Fu_+ + Fu_- = \lambda_+ + \lambda_-$

6. 柯尔劳什 square root law：$\Lambda^{eq} = \Lambda^0 - k\sqrt{c}$

For $A_{\nu_+}B_{\nu_-}$ type：$\lambda_0 = \nu_+\lambda_{0,+} + \nu_-\lambda_{0,-}$

每一种离子传输的电荷量在通过溶液总电量中的分数

7. Transference number 离子迁移数 $t_i = \frac{|z_i|u_i c_i}{\sum_j |z_j| u_j c_j}$

$t_+ = \frac{z_+u_+}{z_+u_+ - z_-u_-}$；支持电解质：溶液电阻↓、电活性离子的迁移↓、电位下降↓ 不参与化学反应，提高κ↑

8. ① Mole fraction $x_i = \frac{n_i}{n_{tot}}$

② Molar concentration 体积摩尔浓度 $c = \frac{n_{solute}}{V_{solution}}$

③ Molal molality 质量… $b = \frac{n_{solute}}{M_{solvent}}$

9. $\mu = \mu^{\ominus} + RT\ln a$. $a = \gamma\frac{b}{b^{\ominus}}$. $\gamma \to 1$ as $b \to 0$

$\mu = \mu^{\ominus} + RT\ln b + RT\ln\gamma = \mu^{ideal} + RT\ln\gamma$

两种离子活度系数平均值

- 平均活度系数 $\gamma_\pm = (\gamma_+\gamma_-)^{\frac{1}{2}}$ $\mu_\pm = \mu^{ideal} + RT\ln\gamma_\pm$

$\left[\log\gamma_\pm = -|z_+z_-|AI^{\frac{1}{2}}.\ I = \frac{1}{2}\sum_i z_i^2\left(\frac{b_i}{b^{\ominus}}\right)\right]$

Chap3 Potentials and Thermodynamics of Cells

一. 相间电位和电极电位

1. Phase 相：空间有一个区域，物理性质均匀

2. 内电势 Φ：将单位正电荷从无穷远处移入M相所做功

外电位 ψ：试验电荷移至距球面 10^{-4}–10^{-5}cm 处

表面电位 χ：单位正电荷穿越偶极子层所做功

$\Phi = \psi + \chi$. 电化学位 $\bar{\mu}_i = \mu_i + nF\Phi$

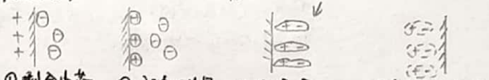
3. Interfacial potential 相间电位

— 两相接触时，两相界面层中存在的电位差.

带电粒子或偶极子在界面层中的不均匀分布.

Reasons：

极性分子在界面溶液一侧定向排列



① 剩余电荷 ② 吸附双电层 ③ 偶极子层 ④ 金属表面电位

⇒ 离子双电层 ⊇ 相间电位 $\left\{\begin{array}{l}\text{外电位差 Volta：}\psi^B - \psi^A\\ \quad\text{直接接触 } \Delta^M\psi^{M^*}\\ \text{内电位差 Galvani：}\Phi^B - \Phi^A\\ \quad\text{电极：}\Delta^M\Phi^S\text{；液体接界 }\Delta^S\Phi^{S^*}\end{array}\right.$

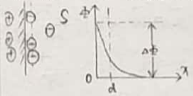
电极电位：电极材料与离子导体的内电位差

4. Electrochemical equilibrium：$\sum_{i=1}^{n} \nu_i \bar{\mu}_i = 0$

5. Electrode potential $\Delta^M\Phi^S = \Phi^M - \Phi^S = \frac{\sum \nu_i \mu_i}{nF} + \frac{\mu_e^M}{F}$

6. Reference electrode 参考电极：电极电位保持恒定

- 绝对电位：电子导电相与离子导电相间的内电位差
- 相对电位：将参比电极与被测电极组成原电池回路. 测得的电池端电压 $E = \varphi - \varphi_R$
- 规定：金属侧剩余正，溶液侧剩余负 $\Rightarrow \Delta^M\Phi^S > 0$

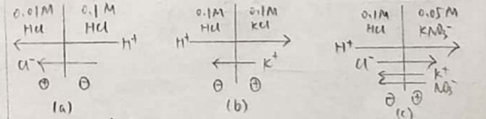


- 标准电位 Standard Hydrogen Electrode

SHE：气体分压 101325 Pa 的 H_2 和活度为1的 H^+ 溶液组成的电极体系

Ex. Daniel cell：$Cu'|Zn|Zn^{2+} \vdots Cu^{2+}|Cu$

7. Liquid junctions 接界电位：相互接触的两个组成不同 or 浓度不同的电解质溶液相间电位. Φ_j



8. 盐桥：连接氧化、还原半电池的实验室装置，液体接界↓

二. 电动势和平衡电极电位

1. Electromotive force (emf)：内电位差的极限值

$dG = \Delta_r G d\xi$, $dw_e = -nFE d\xi \Rightarrow E = -\frac{\Delta_r G}{nF}$

- $E = E^{\ominus} - \frac{RT}{nF}\ln Q_r = E^{\ominus} + \frac{RT}{nF}\ln\frac{a_O}{a_R} \approx E^{\ominus} + \frac{2.3RT}{nF}\log\frac{a_O}{a_R}$
- $E_{half\ cell} = E^{\ominus}_{half\ cell} + \frac{RT}{nF}\ln\frac{a_O}{a_R}$

2. 标准电极电位 $E^{\ominus} = E^{\ominus}_{right} - E^{\ominus}_{(SHE)}$

三. Nernst 方程和标准电极电势的应用

1. 氧化还原能力排序 2. 决定 $E^{\ominus}_{cell} = E^{\ominus}_{right} - E^{\ominus}_{left}$

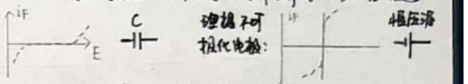
3. 活性因子：$E = E^{\ominus} - \frac{RT}{nF}\ln b^2 - \frac{RT}{nF}\ln\gamma_\pm^2$；$\ln\gamma_\pm = \frac{E^{\ominus} - E}{2RT/F} - \ln b$

4. $\ln K = \frac{nFE^{\ominus}}{RT}$ 5. $\left(\frac{\partial G}{\partial T}\right)_p = -S$. $\left(\frac{\partial G}{\partial p}\right)_T = V \Rightarrow \Delta_r S^{\ominus} = nF\frac{dE^{\ominus}}{dT}$

Chap 4 电极/溶液界面的结构与性质

一. 1. Polarization 极化：有电流流过时电极电位偏离 $\varphi_平$.

Overpotential 过电位：电极电位与平衡电位的差值

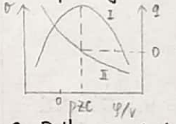


2. Electrical double layer (EDL) 双电层

金属-溶液界面的带电粒子排列与定向偶极子

二. 实验

1. Capillary action 毛细现象 Electrocapillarity 电毛细



I：电毛细

II：表面剩余电荷密度～电位.

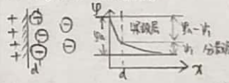
$q = -\left(\frac{\partial\sigma}{\partial\varphi}\right)_{\mu_i}$. (point of zero charge)

2. Differential / integral capacitance

- $C_d = \frac{dq}{d\varphi}$ $C_i = \frac{q}{\Delta\varphi} = \frac{1}{\varphi - \varphi_0}\int_{\varphi_0}^{\varphi} C_d d\varphi$

三. 双电层结构 $\left\{\begin{array}{l}\text{静电作用}\\ \text{扩散作用}\end{array}\right.$ Helmholtz model：$C_{dl} = \frac{\varepsilon\varepsilon_0}{d}$

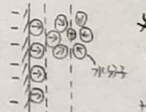
1. Stern model



$\varphi_a = \varphi_紧 + \varphi_分 = (\varphi_a - \psi_1) + \psi_1$

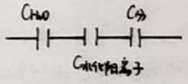
$\frac{1}{C_d} = \frac{d\varphi_a}{dq} = \frac{1}{C_紧} + \frac{1}{C_分}$

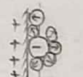
2. 无离子特性吸附时的紧密层结构 $\frac{1}{C_紧} = \frac{1}{C_{H_2O}} + \frac{1}{C_+} = \frac{x_1}{\varepsilon_0\varepsilon_{H_2O}} + \frac{x_2}{\varepsilon\varepsilon_+}$



外紧密层 OHP

$d = x_1 + x_2$





内紧密层 LHP

3. 有特性吸附

4. Adsorption：组分的富集(正吸附)或耗尽(负吸附)

Adsorbate 被吸附物；Adsorbent 吸附剂

- 特性吸附：溶液中的粒子因非静电力作用力发生吸附 $\left\langle\begin{array}{l}\text{物理}\\ \text{化学}\end{array}\right.$

Chap5 Overview of Electrode Processes

一. 1. $\left\{\begin{array}{l}\text{可逆电极：电荷交换与物质传输均平衡}\\ \text{不可逆：有物质变化}\end{array}\right.$

2. $\left\{\begin{array}{l}\text{极化：电流使电子在电极表面积累，电极电位偏离平衡}\\ \text{去极化：电极反应消耗电子，电位} \to \varphi_平\end{array}\right.$

零电荷电位：电极表面剩余电荷为0，不存在双电层的 φ_0；$\varphi_0 > 0$，电极表面+

零标电位差 $\varphi_a = \varphi - \varphi_0$

对于 $O + ne \rightleftharpoons R$

$\varphi_平 = \varphi^0 + \frac{RT}{nF}\ln\frac{a_O}{a_R} = \varphi_0 + \frac{2.3RT}{nF}\log\frac{a_O}{a_R}$

$-\Delta G = nFE$；$\Delta S = nF\left(\frac{\partial E}{\partial T}\right)_p$

$-\Delta H = nFE - nFT\left(\frac{\partial E}{\partial T}\right)_p$

$Q_p = \Delta H$

扫描全能王 创建

Chap 6 Mass Transfer 液相传质

一、三种方式

1. Diffusion 扩散：溶液中的不同区域某组分浓度不同，该组分自发从浓度高区域向低区域移动。（化学位梯度）
2. Migration 电迁移：电解质溶液中的带电粒子在电场作用下沿一定方向移动。
3. Convection 对流：一部分溶液与另一部分溶液间相对流动。

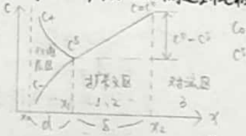
· Nernst-Planck eqn.

$$J_j(x) = -D_j\frac{\partial C_j(x)}{\partial x} - \frac{z_jF}{RT}D_jC_j\frac{\partial \Phi(x)}{\partial x} + C_j v(x)$$

· Fick's first law：稳态扩散流量：$J_i = -D_0\left(\frac{dC_i}{dx}\right)$

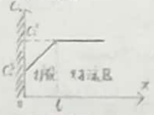
此时，电极反应消耗粒子速率 = 扩散传质输送到电极表面

扩散系数 $D_0 = \frac{\Delta x^2}{2\Delta t}$



二、稳态扩散过程

1. 理想



$j_c = F(-J_{Ag^+}) = FD\frac{C^0_{Ag^+} - C^s_{Ag^+}}{l}$

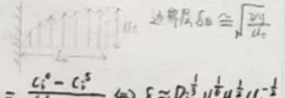
若一般地，$O + ne^- \Rightarrow R$

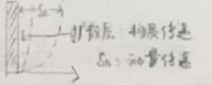
$j = nFD_i\frac{C_i^0 - C_i^s}{l}$ 扩散通量

当 $C_i^s = 0$，$\boxed{\text{极限扩散电流}}$ $j_d = nFD_i\frac{C_i^0}{l}$，完全浓差极化。

· $j = j_d\left(1 - \frac{C_i^s}{C_i^0}\right)$，电极表面附近反应粒子浓度为0。$= nFD_i\left(\frac{dc_i}{dx}\right)_{x=0}$

2. 真实：对流作用下的扩散。





$\delta_{有效} = \frac{C_i^0 - C_i^s}{(dC_i/dx)_{x=0}} \Leftrightarrow \delta \approx D_i^{\frac{1}{3}}\nu^{\frac{1}{6}}y^{\frac{1}{2}}u_0^{-\frac{1}{2}}$

代入得 $j \sim D_i^{\frac{2}{3}}$：对流↓扩散影响。$j = 0.62\,nFD_i^{\frac{2}{3}}\nu^{-\frac{1}{6}}\omega^{\frac{1}{2}}(C_i^0 - C_i^s)$

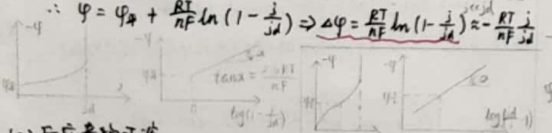
3. 旋转圆盘电极 $u_0 \sim 2\pi n_0 y \Rightarrow \delta = 1.62\,D_i^{\frac{1}{3}}\nu^{\frac{1}{6}}\omega^{-\frac{1}{2}}$

4. 电迁移对扩散的影响 $j = 2FD_+\frac{dc_+}{dx} = 2j_{+,扩散}$

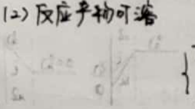
5. 浓差极化的规律 $\varphi = \varphi^0 + \frac{RT}{nF}\ln\frac{\gamma_O C_O^s}{\gamma_R C_R^s}$

(1) 反应产物为独立相：$\gamma_R C_R^0 = 1$，$\gamma_R C_R^s = 1$，$\because C_O^s = C_O^0\left(1 - \frac{j}{j_d}\right)$

$\therefore \varphi = \varphi_平 + \frac{RT}{nF}\ln\left(1 - \frac{j}{j_d}\right) \Rightarrow \Delta\varphi = \frac{RT}{nF}\ln\left(1 - \frac{j}{j_d}\right) \approx -\frac{RT}{nF}\frac{j}{j_d}$（$j \ll j_d$）



(2) 反应产物可溶



$$\begin{cases}\frac{j}{nF} = D_R\left(\frac{C_R^s - C_R^0}{\delta_R}\right),\ C_R^0 = 0 \Rightarrow C_R^s = \frac{j\delta_R}{nFD_R} \\ C_O^s = C_O^0\left(1 - \frac{j}{j_d}\right)\end{cases}$$

当 $j = j_d \cdot \frac{1}{2}$，半波电位 $\varphi_{\frac{1}{2}} = \varphi^0 + \frac{RT}{nF}\ln\frac{\gamma_O\delta_O D_R}{\gamma_R\delta_R D_O} \Rightarrow \varphi = \varphi_{\frac{1}{2}} + \frac{RT}{nF}\ln\frac{j_d - j}{j}$

(3) 扩散控制电极过程 ⇒ j_d：不受 φ 与 T 影响；搅拌↑，j，j_d↑

三、非稳态扩散过程

1. Fick's second law：$\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2}$，$C_i = f(x, t)$。

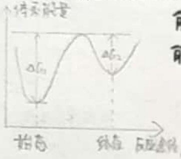
$C_i(x,t) = C_i^0\,\mathrm{erf}\left(\frac{x}{2\sqrt{D_it}}\right)$，真实厚度 $\delta' = 4\sqrt{D_it}$

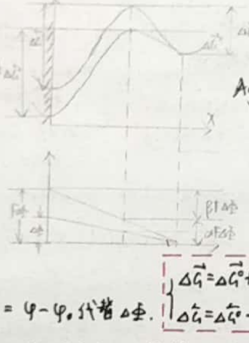
$\Rightarrow j_d = nFD_i\frac{C_i^0}{\sqrt{\pi D_it}}$，$\delta = \sqrt{\pi Dt}$

Chap 7 电子转移步骤动力学：化学反应+电荷转移

一、电极电位对电子转移步骤反应速度的影响

1. 对活化能 Activation energy：反应粒子必须吸收一定能量，才能激发到不稳定的状态，才有可能发生向反应产物的转化。





$Ag^+ + e^- \rightleftharpoons Ag$

(1) 存在界面电场

$\bar{\mu} = \mu + F\Delta\Phi$

$$\begin{cases}\Delta\vec{G} = \Delta\vec{G}^0 + \alpha F\Delta\Phi \\ \Delta\overleftarrow{G} = \Delta\overleftarrow{G}^0 - \beta F\Delta\Phi\end{cases}$$

$$\begin{cases}\alpha：还原 \\ \beta：氧化\end{cases}$$ 传递系数

一般地：φ_a 零标电位 $= \varphi - \varphi_0$ 代替 $\Delta\Phi$。

$$\begin{cases}\Delta\vec{G} = \Delta\vec{G}^0 + \alpha nF\varphi \\ \Delta\overleftarrow{G} = \Delta\overleftarrow{G}^0 - \beta nF\varphi\end{cases}$$

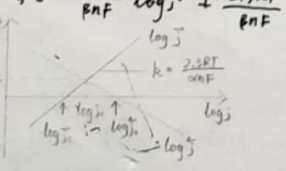
$\Delta\vec{G}^0 / \Delta\overleftarrow{G}^0$：所选用的电位坐标系零点时还原/氧化反应活化能。

2. 对反应速度 $O + e = R$ $\quad v = kc\exp\left(-\frac{\Delta G}{RT}\right)$，$j = nFv$

$$\begin{cases}\vec{j} = nF\vec{k}C_O\exp\left(-\frac{\Delta\vec{G}}{RT}\right) \\ \overleftarrow{j} = nF\overleftarrow{k}C_R\exp\left(-\frac{\Delta\overleftarrow{G}}{RT}\right)\end{cases}$$ 令 $$\begin{cases}\vec{K} = \vec{k}\exp\left(-\frac{\Delta\vec{G}^0}{RT}\right) \\ \overleftarrow{K} = \overleftarrow{k}\exp\left(-\frac{\Delta\overleftarrow{G}^0}{RT}\right)\end{cases}$$

可得 $$\begin{cases}\vec{j}^0 = F\vec{K}C_O \\ \overleftarrow{j}^0 = F\overleftarrow{K}C_R\end{cases}$$，$$\begin{cases}\vec{j} = \vec{j}^0\exp\left(-\frac{\alpha nF\varphi}{RT}\right) \\ \overleftarrow{j} = \overleftarrow{j}^0\exp\left(\frac{\beta nF\varphi}{RT}\right)\end{cases}$$

$$\begin{cases}\varphi = \frac{2.3RT}{\alpha nF}\log\vec{j}^0 - \frac{2.3RT}{\alpha nF}\log\vec{j} \\ \varphi = -\frac{2.3RT}{\beta nF}\log\overleftarrow{j}^0 + \frac{2.3RT}{\beta nF}\log\overleftarrow{j}\end{cases}$$



二、基本动力学参数

1. 交换电流密度 j_0：无外电流，但微观有物质交换

$$\begin{cases}j_0 = \vec{j} = nF\vec{K}C_O\exp\left(-\frac{\alpha nF\varphi_平}{RT}\right) \\ j_0 = \overleftarrow{j} = nF\overleftarrow{K}C_R\exp\left(\frac{\beta nF\varphi_平}{RT}\right)\end{cases}$$

· 与 k，C，电极材料相关

对于 $O + e \rightleftharpoons R$，$\alpha + \beta = 1$（单电子反应）

代入得 $\varphi_平 = \frac{RT}{F}\ln\frac{\vec{K}}{\overleftarrow{K}} + \frac{RT}{F}\ln\frac{C_O}{C_R} = \varphi^{0\prime} + \frac{RT}{F}\ln\frac{C_O}{C_R}$

且 $\varphi^{0\prime} = \varphi^0 + \frac{RT}{F}\ln\frac{\gamma_O}{\gamma_R}$

$\therefore \vec{j} = F\vec{K}C_O\exp\left[-\frac{\alpha F}{RT}(\varphi_平 + \Delta\varphi)\right] = j^0\exp\left(-\frac{\alpha F}{RT}\Delta\varphi\right)$

净反应速度 $j_净 = j^0\left[\exp\left(-\frac{\alpha F}{RT}\Delta\varphi\right) - \exp\left(\frac{\beta F}{RT}\Delta\varphi\right)\right]$

若 $\alpha \approx \beta \approx 0.5$，可认为 $j_净 \propto j^0$；j^0 大 ⇒ 去极化作用强。

2、电极反应速度常数 K。 $O + e \rightleftharpoons R$

当处于平衡电极电位 $\varphi^{0\prime}$ 时，$C_O = C_R$。

$K = \vec{K}\exp\left(-\frac{\alpha F\varphi^{0\prime}}{RT}\right) = \overleftarrow{K}\exp\left(\frac{\beta F\varphi^{0\prime}}{RT}\right)$

↓ φ为标准电极电位和反应粒子浓度为单位浓度时电极反应绝对速度。

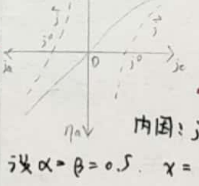
$\Rightarrow j^0 = \vec{j} = FKC_O\exp\left[-\frac{\alpha F}{nF}(\varphi - \varphi^{0\prime})\right]$（平衡时）

$\Rightarrow j^0 = FKC_O^{\beta}C_R^{\alpha}$

三、稳态电化学极化规律

1、Tafel law $\eta = a + b\log i$

2、BV方程 电极反应 $j = j_净$，j — 外电流/极化电流



阴 $j_c = \vec{j} - \overleftarrow{j} = j^0\left[\exp\left(\frac{\alpha F}{RT}\eta_c\right) - \exp\left(-\frac{\beta F}{RT}\eta_c\right)\right]$

阳 $j_a = \overleftarrow{j} - \vec{j} = \cdots$

· 过电位是电极（净）反应的推动力

内因：j_0，外因：j

设 $\alpha = \beta = 0.5$，$x = -\frac{F}{2RT}\Delta\varphi$。⇒ $j = 2j^0\sinh x$（双曲）

3、高电位下 $|\Delta\varphi|$ 很大，考虑阴极极化

$\because \exp\left(\frac{\alpha F}{RT}\eta_c\right) \gg \exp\left(-\frac{\beta F}{RT}\eta_c\right) \Rightarrow j_c \approx j^0\exp\left(\frac{\alpha F}{RT}\eta_c\right)$

$$\begin{cases}\eta_c = -\frac{2.3RT}{\alpha F}\log j^0 + \frac{2.3RT}{\alpha F}\log j_c \\ \eta_a = -\frac{2.3RT}{\beta F}\log j^0 + \frac{2.3RT}{\beta F}\log j_a\end{cases}$$

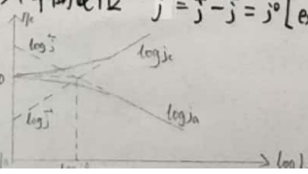
4、低电位下 $j \approx j^0\left[1 - \frac{\alpha F\Delta\varphi}{RT} - \left(1 + \frac{\beta F\Delta\varphi}{RT}\right)\right]$

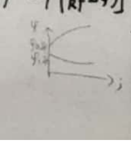
$\alpha + \beta = 1 \Rightarrow j \approx -\frac{F\Delta\varphi}{RT}j^0$，$\Delta\varphi \approx -\frac{RT}{Fj_0} \cdot j$

· Linear Law：$\eta = \bar{\omega}j \Rightarrow \bar{\omega} = \frac{RT}{Fj_0}$

· 极化/反应电阻 $R_r = \frac{RT}{Fj_0}$

5、中间电位 $j = \vec{j} - \overleftarrow{j} = j^0\left[\exp\left(\frac{\alpha F}{RT}\Delta\varphi\right) - \exp\left(\frac{\beta F}{RT}\Delta\varphi\right)\right]$





四、混合控制

· 浓差极化：主要影响电极表面反应粒子浓度 C^s

$$\begin{cases}\vec{j} = nFK_cC_O^s\exp\left(-\frac{\vec{\alpha}F}{RT}\varphi\right) = j^0\frac{C_O^s}{C_O^0}\exp\left(-\frac{\vec{\alpha}F}{RT}\Delta\varphi\right) \\ \overleftarrow{j} = nFK_aC_R^s\exp\left(-\frac{\overleftarrow{\alpha}F}{RT}\varphi\right) = j^0\frac{C_R^s}{C_R^0}\exp\left(-\frac{\overleftarrow{\alpha}F}{RT}\Delta\varphi\right)\end{cases}$$

$\therefore j = j^0\left[\frac{C_O^s}{C_O^0}\exp\left(-\frac{\vec{\alpha}F}{RT}\Delta\varphi\right) - \frac{C_R^s}{C_R^0}\exp\left(\frac{\overleftarrow{\alpha}F}{RT}\Delta\varphi\right)\right]$

1、高电位下 $|j| \gg j^0$，$j_c \approx \vec{j} = j^0\frac{C_O^s}{C_O^0}\exp\left(-\frac{\vec{\alpha}F}{RT}\Delta\varphi\right)$

由 $C_O^s = C_O^0\left(1 - \frac{j_c}{j_d}\right)$ 得 $j_c = j^0\left(1 - \frac{j_c}{j_d}\right)\exp\left(-\frac{\vec{\alpha}F}{RT}\eta_c\right)$

$\therefore \eta_c = \underbrace{\frac{RT}{\vec{\alpha}F}\ln\frac{j_c}{j^0}}_{电化学极化} + \underbrace{\frac{RT}{\vec{\alpha}F}\ln\frac{j_d}{j_d - j_c}}_{浓差/扩散过电位}$ → 浓差极化极化参数

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1. Galvanic cell 原电池：电化学体系中的两个电极与外电路负载接通后，能自发地将电流送到外电路中做功的体系。

Electrolytic cell 电解池：与外电源组成回路，强迫电流在电化学体系中通过并促使电化学反应发生的体系。

2. Cathode 阴极：还原反应；Anode 阳极：氧化反应

3. $\left(\frac{\partial G}{\partial n_j}\right)_{T,p,n'}$：Partial derivative；

· Lonic strength：A measure of the concentration of ions in the solution

4. 最大效率：热机 $\eta = 1 - \frac{T_c}{T_h}$；Fuel cell $\eta = \frac{\Delta G}{\Delta H} = 1 - \frac{T\Delta S}{\Delta H}$

	ΔH	Δ
H_2	<0	<0
CO	<0	<
CH_4	<0	~

5. Gibbs energy：一个热力学系统中，系统减少的内能中可以转化为对外做功的部分；molar…：在标准状态下，判断反应进行方向的

6. 化学势：表征系统与環境 or 相与相 or 系统组元之间粒子转移的趋势

逸度：相同条件下具有相同化学势的理想气体的压强。

7. $G = H - TS$，$H = U + PV$，$U = Q + W \Rightarrow dG = d(U + W) + d(pV) - d(TS)$

$dG = \delta Q + \delta W_{体} + \delta W_{非体} + d(pV) - d(TS) \xlongequal{可逆} TdS - pdV + \delta W_{非体} + pdV + Vdp - \ldots$

$= Vdp - SdT + \delta W_{非体}$，Isothermic 等温 and isobaric 等压：$-dG = -W_{非体}$

8、$dS_{iso} = dS_{sys} + dS_{env} \geq 0 \Rightarrow dS_{sys} \geq -dS_{env} \Rightarrow \delta Q_{sys} \leq TdS_{sys}$

$\therefore dG \leq Vdp - SdT + \delta W_{非体}$，等 T，p ⇒ $-dG \geq -W_{非体}$

9、$dG = RT\ln f$，$f = \Phi p$；等 T 时，$dG = Vdp \Rightarrow RT\,d\ln\left(\frac{f}{p}\right) = \left(V - \frac{RT}{p}\right)dp$

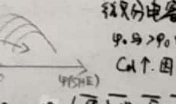
$\Rightarrow d\ln\Phi = \frac{1}{RT}\left(V - \frac{RT}{p}\right)dp$，$pV = RTZ \Rightarrow \ln\Phi = \int_0^p\frac{Z-1}{p}dp$

10、Kohlrausch's 独立运动：无限稀释，离子迁移不受其它离子影响，独立移

11、$t_i = \frac{j_i}{\sum j_i}$，$j_i = z_iFC_i \cdot v_i$。

12、毛细曲线

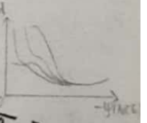
阴离子吸附改变了电双层结构



微分电容曲线

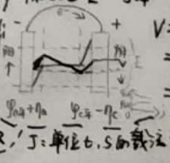
φ_0 与 $>\varphi_0$ 电位内，

Cd↑，因 $\delta_{分散}$↓。



13、阴极极化：$\eta_c = \varphi_平 - \varphi_c$，阳极极化：$\eta_a = \varphi_a - \varphi_平$

原电池 $E = \varphi_{c平} - \varphi_{a平}$

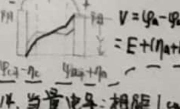


$V = \varphi_c - \varphi_a - IR$

$= (\varphi_{c平} - \eta_c) - (\varphi_{a平} + \eta_a) - IR$

$= E - (\eta_c + \eta_a) - IR$

电解池 $E = \varphi_{a平} - \varphi_{c平}$



$V = \varphi_a - \varphi_c + IR$

$= E + (\eta_a + \eta_c) + IR$

J：单位 t，S 的载流子量。

14、当量电导：相距 1 cm 的 S 相对的平行板电极间，含有 1 g 当量电解质溶液所具有的电导。

15、浓差电池 $$\begin{cases}对+离子可逆：E = 2t_-\frac{RT}{F}\ln\frac{a_\pm'}{a_\pm} \\ 对-离子可逆：E = 2t_+\frac{RT}{F}\ln\frac{a_\pm}{a_\pm'}\end{cases}$$

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